

Quantifiers

Date

In the discussion of logic, some very important statements contain quantifiers.

e.g. *

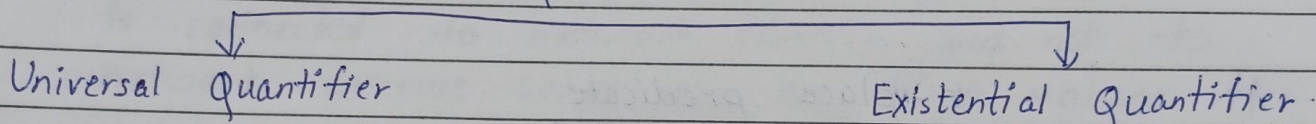
* Some people are honest.

* No woman is a player.

* All Americans are rich.

The words some, no and all are known as quantifiers.

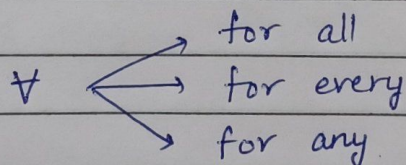
Quantifiers



Universal Quantifier :-

Let p be a statement. We define the symbol $\forall x P(x)$ to mean that for every value of x in the given set, the statement $P(x)$ is true.

The symbol $\forall$ is called the universal quantifier.



e.g.:- The statement "for all natural numbers, $n+4 > 3$ " can be expressed as $\forall x P(x)$, where x belongs to the set N of natural numbers and p is the statement ' $n+4 > 3$ '.

Existential Quantifier :-

Let p be a statement. We define the symbol $\exists x P(x)$ to mean that for one or more elements x of a certain set, the proposition p is true. The symbol $\exists$ is called existential quantifier.

$\exists$ $\begin{cases} \text{there exists} \\ \text{for at least one} \\ \text{for some.} \end{cases}$

e.g.: (1) The statement "there exists a number x such that $x^2 - 4x = 16$ may be written as $\exists x (x^2 - 4x = 16)$.

(2) If $P(x)$ was the statement $x > 89$, and your data set included test scores of 65, 72, 85, 88 and 95, what would be the existential quantification of $P(x)$?

$\exists x P(x) \rightarrow \text{True}$

Statement	When true	When false
$\forall x P(x)$	$P(x)$ is true for every x	There is an x for which $P(x)$ is false
$\exists x P(x)$	There is an x for which $P(x)$ is True	$P(x)$ is false for every x

Free and Bound variables

A variable in a predicate is bound by a Quantifier.
A variable is free if it is not bounded.

eg:- $\forall x P(x, y)$ and $\exists x P(x, y)$

In both cases, x is a bound variable and y is a free variable.

* An expression with zero free variables is a actual (bona-fide) proposition.

* An expression with one or more free variables is still only a predicate. [$\exists x P(x, y)$]

Ques:- Let $P(x)$: x is even number and the universe of discourse for x is the set $\{1, 2, 3, 4\}$.

(a) Find the truth value of $\forall x P(x)$.

(b) Find the truth value of $\exists x P(x)$.

Precedence of Quantifiers :-

The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators in the propositional calculus.

e.g:-

$$\forall x P(x) \vee Q(x) \neq \forall x (P(x) \vee Q(x))$$

$$\forall x P(x) \vee Q(x) = (\forall x P(x)) \vee Q(x)$$

Negation of Quantifiers:-

Every Politician is clever

$P(x)$: x is clever.

UD is set of politicians.

Symbolized as $\forall x P(x)$

Negation :-

It is not the case that every politician is clever
or

There is a politician who is not clever.

$\exists x \neg P(x)$.

Date

Write the negation of following

(a) All states in India are highly populated.

(b) Some states in India are highly populated.

Laws of equivalence :-

$\neg \forall x P(x) \equiv \exists x \neg P(x)$
$\neg \exists x P(x) \equiv \forall x \neg P(x)$

Proof :-

$$\begin{aligned} & \neg \forall x P(x) \\ & \equiv \neg (P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge \dots \wedge P(x_n)) \\ & \equiv \neg P(x_1) \vee \neg P(x_2) \vee \neg P(x_3) \vee \dots \vee \neg P(x_n) \\ & \equiv \exists x \neg P(x) \end{aligned}$$

Negation of quantified statements with more than one variable.

$$\begin{aligned} 1* \quad & \neg [\forall x \forall y P(x, y)] \\ & \equiv \exists x [\neg \forall y P(x, y)] \\ & \equiv \exists x \exists y [\neg P(x, y)] \end{aligned}$$

$$\begin{aligned} 2* \quad & \neg [\exists x \exists y \neg P(x, y) \wedge \forall x \forall y Q(x, y)] \\ & \equiv \forall x \forall y P(x, y) \vee \exists x \exists y \neg Q(x, y) \end{aligned}$$

$$\begin{aligned} 3* \quad & \neg [\forall x \exists y P(x, y)] \\ & \equiv \exists x [\neg \exists y P(x, y)] \\ & \equiv \exists x \forall y [\neg P(x, y)] \end{aligned}$$

Quantifier $\longrightarrow$ Predicate

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

$$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$$

where $\{x_1, x_2, \dots, x_n\}$ is the Universe of disco